PROJECT DESCRIPTION REPORT

1. Project Title and Business Idea

Project Title: Hotel Booking System

Business Idea:

The Hotel Booking System is a proposed information system designed to help a hotel manage guest information, room details, reservations, and payments. The system will organize hotel booking information and make it easier to manage reservations and room availability.

2. Problem Statement

Some hotels manage guest information, room reservations, and payment records using paper documents or separate records. This can make it difficult to retrieve information, track room availability, manage reservations, and maintain accurate payment records. These challenges may lead to booking errors, lost information, and delays in providing services to guests.

3. Purpose of the Proposed System

The purpose of the proposed Hotel Booking System is to provide a clear and organized way to manage hotel bookings and related information. It aims to simplify reservation management, improve the accuracy of records, track room availability, and organize payment information.

The project will use a UML class diagram to represent the main classes of the system and the relationships between them.

4. Project Scope

The scope of the Hotel Booking System is to model how a hotel manages guest information, room details, reservations, and payments. The project focuses on identifying the main classes, their attributes and methods, and the relationships between them using a UML class diagram. It covers the basic activities involved in reserving hotel rooms and maintaining booking records. The project is limited to system analysis and UML modelling; it does not involve developing a complete working application.

Functions Covered

The proposed system will include the following functions:

- Managing guest information.
- Storing hotel room details, room types, prices, and availability status.
- Creating and managing room reservations.

- Recording booking dates and booking status.
- Recording payment details associated with bookings.
- Calculating the total booking cost based on the room price and length of stay.

Functions Excluded

The proposed system will not cover:

- Online payment gateway integration.
- Housekeeping and room cleaning management.
- Employee payroll and staff management.
- Restaurant and food ordering services.
- A complete, fully functional hotel booking application.

The project focuses on analysing the proposed system and designing its UML class diagram.

5. Main Classes and Their Responsibilities

The proposed system consists of four main classes:

1. Guest

This class stores guest information, including guest ID, full name, and phone number. It supports creating bookings and updating guest details.

2. Room

This class stores room information, including room number, room type, price per night, and availability status. It supports checking room availability and updating room status.

3. Booking

This class stores reservation information, including booking ID, check-in date, check-out date, and booking status. It supports creating bookings, cancelling reservations, and calculating the total booking cost.

4. Payment

This class stores payment information, including payment ID, amount, payment date, and payment method. It supports recording payments and issuing receipts.

6. Brief Description of System Requirements

Functional Requirements

The system should be able to:

1. Store and update guest information.
2. Store room details and availability status.
3. Create, view, and cancel bookings.
4. Record check-in and check-out dates.
5. Calculate the total cost of a booking.
6. Record payment details and associate payments with bookings.

Non-Functional Requirements

The proposed system should meet the following requirements:

1. Accuracy: Information about guests, rooms, bookings, and payments should be consistent and correct.
2. Usability: The system should be easy to understand and use.
3. Reliability: Records should be organized consistently to reduce errors and information loss.
4. Security: Guest and payment information should be protected from unauthorized access.
5. Maintainability: The system design should be clear enough to support future improvements.